

Annual Review

Grid2Op

Proposal for advancement to Incubation Stage

What is Grid2Op?

Open-source power grid simulation framework

Grid2Op provides a simulation environment for running fast and reproducible "experiments" on power grids.

Key capabilities

- Sequential decision-making on realistic grid scenarios
- Pluggable "backends" ([pandapower](#), [lightsim2grid](#), [pypowsybl](#), [Power Grid Model](#) ?...)
- Pluggable "grid controllers" (machine learning, reinforcement learning, heuristic, optimization based, ...)
- N-1 security analysis, forecasting, multi-agent support
- Used in the L2RPN competition series

Example of "experiments"

- Evaluating the performance of a controller in various scenarios (generation and demand)
- Benchmarking performance of "solvers" facing the same scenarios
- Generating data varying generation, demand, solver, topology etc.
- ...

1.12

Current grid2op version

505

Commits (past year, grid2op / lightsim2grid)

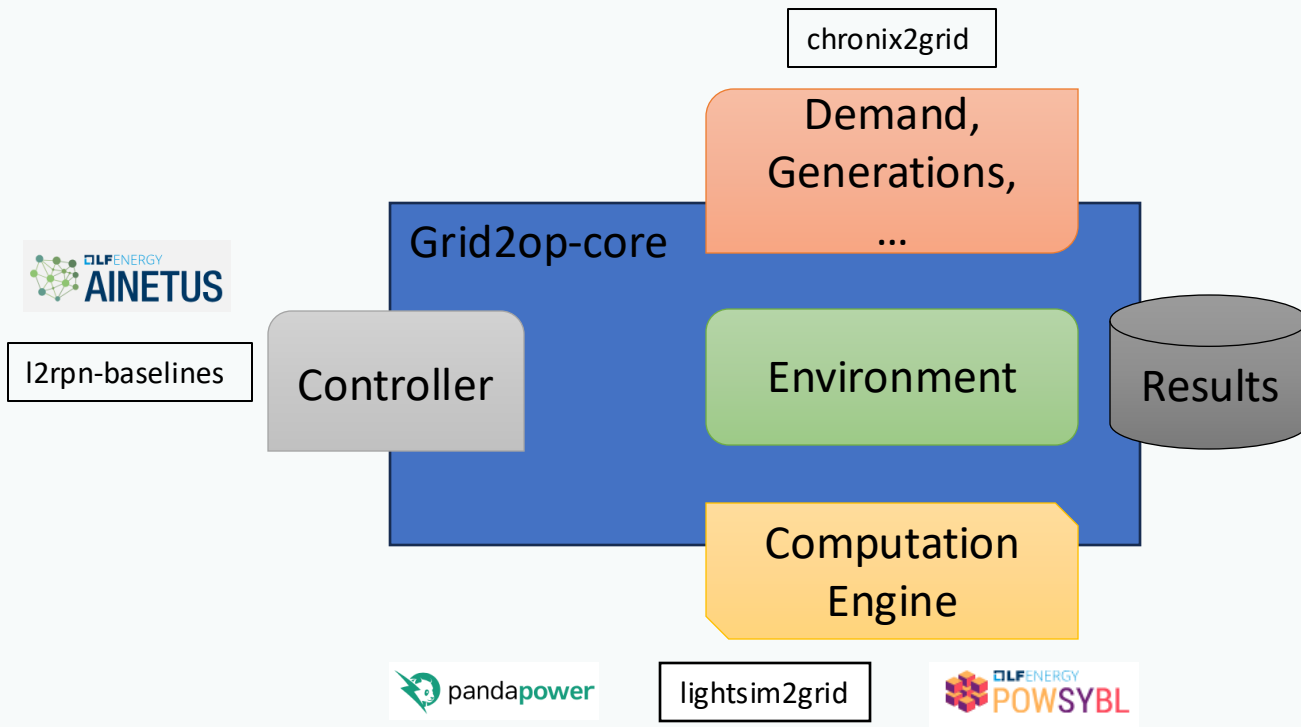
4+

Contributing orgs

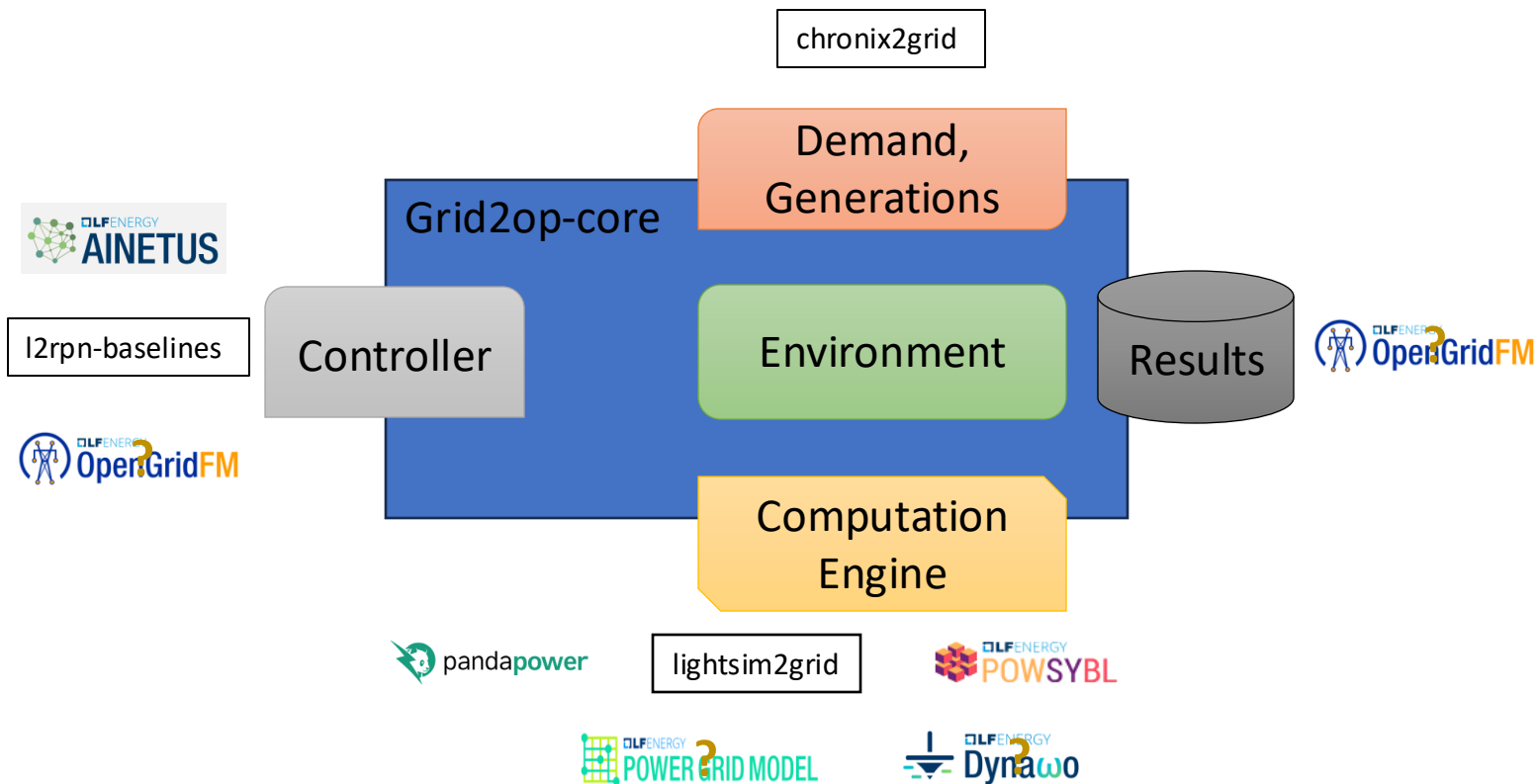
2019

Project started

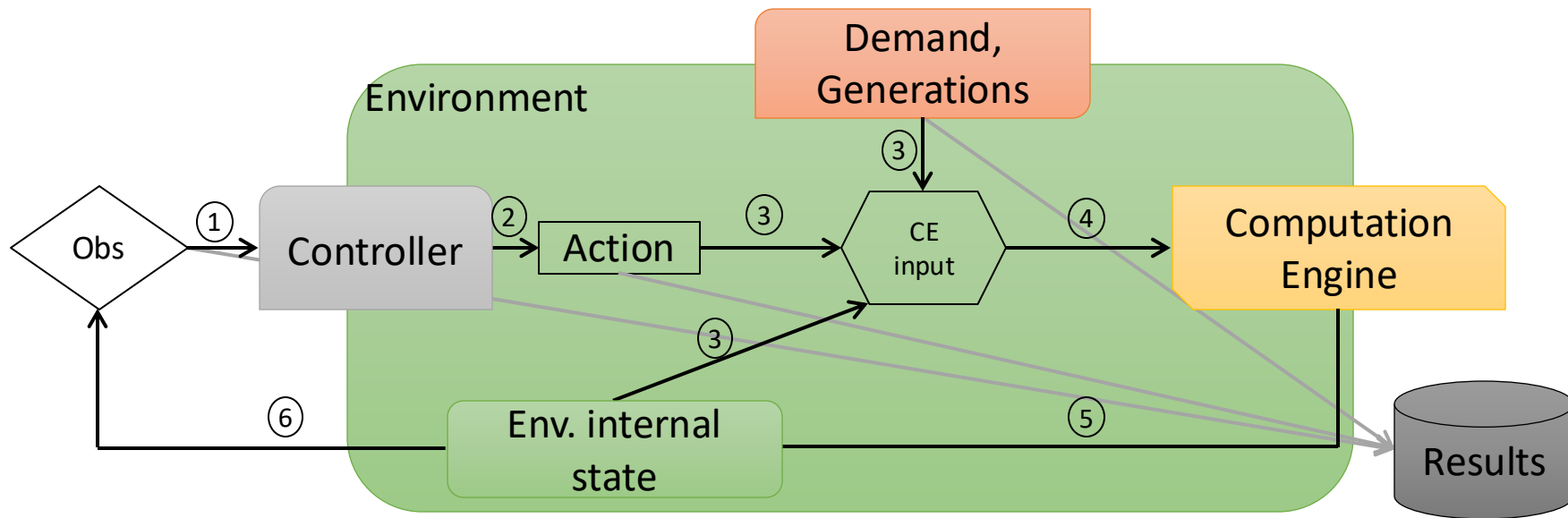
Overall structure



Possible integration in LF Energy landscape



Overview of a “grid2op step”



Ecosystem Structure

The Grid2Op organization uses a two-tier package model, endorsed by LF Energy

Core Packages

Well-maintained, well-documented, extensively tested.
Subject to full governance process and OpenSSF requirements.

grid2op

lightsim2grid

pypowsybl2grid *(candidate for Core promotion)*

Example / Companion Packages

Experimental or companion contributions. Lighter governance. May include:

- L2RPN competition baselines
- UX / HMI implementations
- Data generation (e.g. chronix2grid)
- Research paper contributions

Archive policy: unmaintained Example packages can be archived by TSC vote, with explicit warnings to users and last known compatible Core versions documented.

Incubation Criteria — Status

- ✓ [LICENSE](#) file (MPL v2.0)
- ✓ [README](#) welcoming new contributors
- ✓ [CONTRIBUTING.md](#) with process details
- ✓ CODEOWNERS / [COMMITTERS](#) file
- ✓ [CODE_OF_CONDUCT](#) (Linux Foundation)
- ✓ [RELEASE.md](#) — cadence & criteria documented
- ✓ [GOVERNANCE.md](#) — technical governance
- ✓ [SUPPORT](#) file
- ✓ Technical Charter signed / trademarks transferred
- ✓ [OpenSSF Best Practices Badge](#) — Passing level
- ✓ License scan completed, minor issues fixed in 1.12.5 (in the docs)
- ✓ [Project mission & scope defined](#)

Roadmap defined ✓ | Architecture overview documented ✓ | ArchiMate diagram: documented in slides 3-5

Contributions — Past 12 Months (grid2op + lightsim2grid)

505

Total commits

5

Individual contributors

4

Contributing organizations

Contributors

Popularity

Development

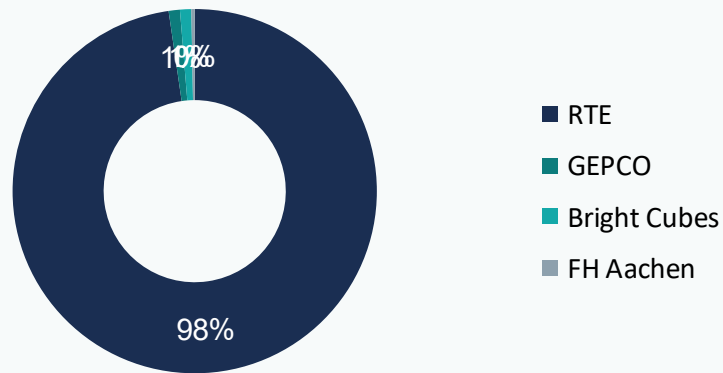
Security & Best practices

Contributors

Contributor	Commits	%
Benjamin Donnot (RTE)	771	97%
Ahmad Furqan (GEPCO)	7	1%
Martijn Govers (Bright Cubes)	4	1%
Mattholomew	3	< 1%
Florian Maurer (FH Aachen)	1	< 1%

*Benjamin Donnot appears under 3 identities in LFX; consolidated here.

Contributions by Organization



Organizations Contributing & Using Grid2Op



RTE — Réseau de Transport d'Électricité

Creator & primary maintainer

French TSO. Developed Grid2Op to power the L2RPN competition series and internal research on AI-assisted grid dispatching.



Fraunhofer IEE

Active user & contributor

Developed CurriculumAgent (open-source RL agent) on top of Grid2Op. Uses the framework for power systems research.



KTH Royal Institute of Technology

Active user & contributor

Uses Grid2Op for academic research on reinforcement learning for power grid control.



AI4RealNet (EU project)

Active user

European research project using Grid2Op as the simulation environment for its power grid use case.



GEPCO (Gujranwala Electric Power Company)

Contributor

Recent code contributor (7 commits in past year).



Bright Cubes

Contributor

Contributing to Grid2Op and lightsim2grid (4 commits).

Organizations Contributing & Using Grid2Op



Decent activity and popularity



Contributor Dependency

1 contributor accounts for 51%+ of contributions - This project is dependent on a single contributor, leading to a high risk if that individual becomes unavailable.



Organization Dependency

1 organization accounts for 51%+ of contributions - This project is highly dependent on a single organization, indicating a lack of diverse organizational support.

Bus factor == 1

Key Achievements — April 2025 to May 2026

Element detachment & load shedding (v1.11.0)

Agents can now disconnect loads, generators, and storage units — a long-requested feature enabling realistic emergency scenarios and expanding the action space for RL research.

Modern Python stack compatibility (v1.12.0)

Full support for numpy 2, scipy ≥ 1.14 , Python 3.13, and pandapower 3. Keeps Grid2Op aligned with the scientific Python ecosystem used by the research community.

Significant performance improvements (v1.12.1)

Faster computation across the board: cached ambiguity checks, optimised 'do nothing' paths, fewer unnecessary copies. Critical for RL training loops calling `env.step()` millions of times.

Security hardening & governance maturity (v1.11–1.12.5)

CodeQL integration, SonarCloud, pre-commit secret-scanning hooks, `SECRETS_MANAGEMENT.md`, removal of `rest_server` for security. OpenSSF Best Practices badge achieved.

Growing academic adoption — 2025/2026 publications

Grid2Op cited in RL2Grid (ICLR benchmark), Springer ECML-PKDD 2026 chapter on topology control, ACM e-Energy 2025 multi-agent RL paper, and arXiv distributed RL paper (Sep 2025).

Areas Where the Project Could Use Help

Community Structure & Onboarding

- Lower the barrier to first contribution: clearer 'good first issue' labels, contributor mentoring
- Reduce the perceived intimidation of a physics-heavy, large codebase for new developers
- Structure the TSC to actively invite and retain contributors from outside RTE

Promotion & Outreach

- Increase visibility outside the RL research community (utilities, energy-sector developers)
- Leverage synergies with LFE sibling projects: OpenGridFM, PowerGridModel, Dynawo
- Support participation in energy-sector conferences beyond IEEE competitions

Branding & Project Identity

- Cleaner project website — current presence is GitHub-only
- Clearer positioning vs. other power grid simulation frameworks
- Guidance from LFE on how to build a project brand within the foundation ecosystem

Feedback on Working with LF Energy

What works well

“The diagnostic tooling LFE provides — LFX Insights, OpenSSF badge, license scans — genuinely identified weaknesses in the project that we have since addressed. This made Grid2Op a more mature, more secure, and more trustworthy open-source project.

Quality and speed of responses from LFE staff has been excellent throughout.

Suggestions for improvement

“More guidance on community-building practices specific to scientific/research open-source projects would be valuable — the typical contributor profile here is an academic or a domain expert, not a professional open-source developer.

Examples of how other LFE projects successfully grew their contributor base beyond a single organization would be very helpful.

Roadmap

Organisation-level roadmap —

Governance, two-tier package structure, and cross-cutting commitments are fully documented. Published at github.com/Grid2Op (currently in the grid2op repo, pending migration to the organisation level)

Core repository roadmaps —

Detailed roadmaps for grid2op and lightsim2grid are being drafted, targeting publication before the next TAC cycle.

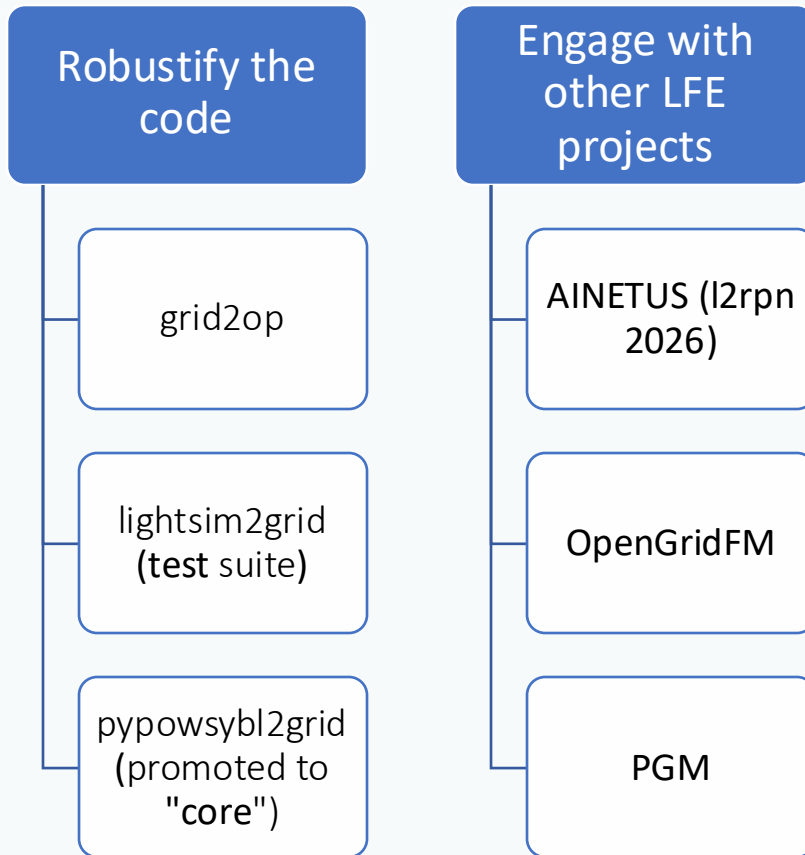
Focus of 2026 → September 2027

3 key areas identified as a priority

- **Speed** — *eg.* faster `env.step()` and parallel execution for large-scale RL training
- **Realism** — *eg.* better modelling of switches, transformers, HVDC, and shunts
- **Research friendliness** — *eg.* cleaner API, better documentation, easier backend integration

Question for the TAC: Grid2Op has active research users (KTH, Fraunhofer, AI4RealNet). We will seek TAC guidance on what constitutes a qualifying production user for Early Adoption, and how other LFE projects in a similar position have approached this.

Roadmap June → August 2026



Making the code more robust

- Plan to apply to google [oss-fuzz](#)
- Set-up different analysis tools:
 - Codacy, sonarqube
 - Github codeQL
 - Openssf scorecard
- Lightsim2grid is mainly cpp (not memory safe). Any recommendations on best practices for the continuous integration of such codes ?

Roadmap Sep. 2026 → Sep. 2027

grid2op-org

- OpenGridFM
- PGM

grid2op

- Switches
- Action on transformers

lightsim2grid

- Modularize cpp code
- HVDC

pypowsybl2grid

- Shunts
- Storage units
- Detailed topology

Candidate for a security analysis audit

Grid2op-core is pure python, lightsim2grid is (almost entirely) cpp, any security check is worth doing.

Thank you

We respectfully request the TAC's affirmative vote to advance Grid2Op from Sandbox to Incubation stage.

GitHub: github.com/Grid2Op/grid2op

Documentation: grid2op.readthedocs.io

LFX Insights: insights.lfx.linuxfoundation.org (grid2op)

Mailing list / discussions: github.com/Grid2Op/grid2op/discussions